

These components are inclusive of technologies like cloud computing, virtualisation, green networking, and re-designing of data centres (Figure 2).

The major components are described below.

Green data centres: The enormous amount of data generated every day needs to be stored and processed in huge data centres. Thousands of computing systems in these data centres are established with gigantic cooling infrastructures because these systems have high power consumption and generate a large amount of heat, thereby consuming enormous amount of energy per day. In fact, according to some sources, if we try to group all these data centres to form a single country, the country formed may stand fifth in the list of countries that consume most energy in the world.

And there is no doubt that these scary figures are just going to get worse with time in coming years. Hence, it is quite important to devise new technologies that are much greener and based completely on different mechanisms.

It is estimated that there will be nearly ten times increase in data creation by 2025. But until and unless we bring in greener architectures, this would result in a huge generation of greenhouse gases and carbon footprints. The data centres would be considered green only when they fulfil the following paradigms:

- Minimisation of carbon footprints of data centre buildings
- E-waste from these data centres is recycled
- Building material and paints produce less emission on heat generation
- Hybrid and electric vehicles are used at these data centres
- There is use of evaporative cooling, photovoltaic technology, and heat pumps

Innovation in the redesigning of data centres has begun in some

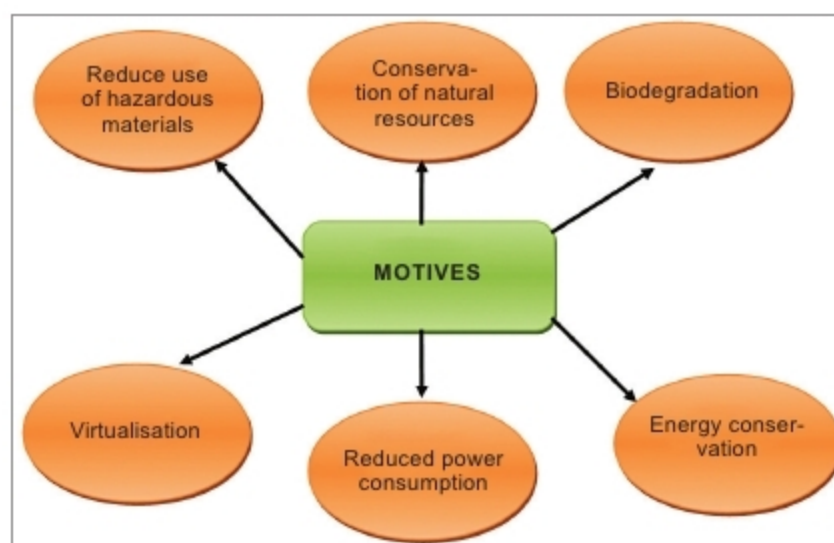


Figure 1: Motives of green IT

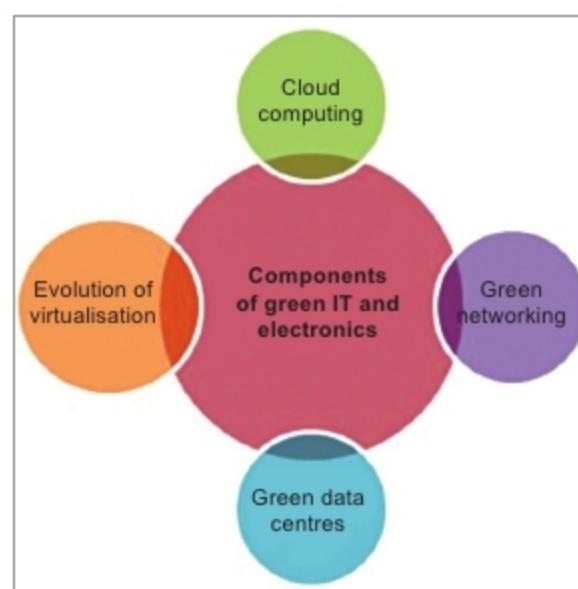


Figure 2: Components of green IT

countries. Companies like SuperMicro have introduced resource-saving architecture, which minimises e-waste, pollution, power consumption, and much more. Intel has come up with energy-efficient data centres that reduce power consumption.

Green networking: When we use green and energy-efficient devices in order to minimise resource utilisation in computer networking, it is called green networking. It deals with energy-efficient communication strategies with respect to the environment. For instance, we can utilise energy on the Internet by setting routers to sleep mode in low traffic periods and then utilise the same router for multiple purposes. Green networking can be practised in the following ways:

- Reducing remote communication/video conferencing
- Making servers more strong and resilient
- Employing system management

- Using greener or energy-efficient products

Cloud computing:

Cloud computing is one of the hottest trends in the IT market these days. It delivers on-demand computing services—from applications to storage and processing power—typically over the Internet and on a pay-as-you-go basis.

Evolution in cloud computing in coming years will lead to the utmost resource utilisation by resource sharing. Instead of buying multiple systems for different operating systems, one can buy just one operating system (OS), set it up online on the cloud, and operate it liberally. This also helps in reducing the amount of e-waste generated.

Virtualisation: Virtualisation has come from its root word virtual, which means 'not real.' The concept of virtualisation puts forth a mirage of resources in front of IT users.

Virtualisation in system memory brings in the utilisation of memories, showing them to be much larger than these actually are. The same concept is used in cloud computing and other similar technologies.

A virtual box installed on a computer system can do wonders these days. One can buy resources online from vendors and install a virtual box, irrespective of its compatibility with the system and thereby enjoy all sorts of services. Virtualisation indeed is one of the most important pillars of green computing.

Green computing

To promote green computing concept the following approaches are employed:

- Minimising energy consumption in an eco-friendly manner
- Safe disposal of e-waste
- Energy-efficient design of electronics and computer products
- Minimising waste during the